# CFD Analysis of a NACA Airfoil

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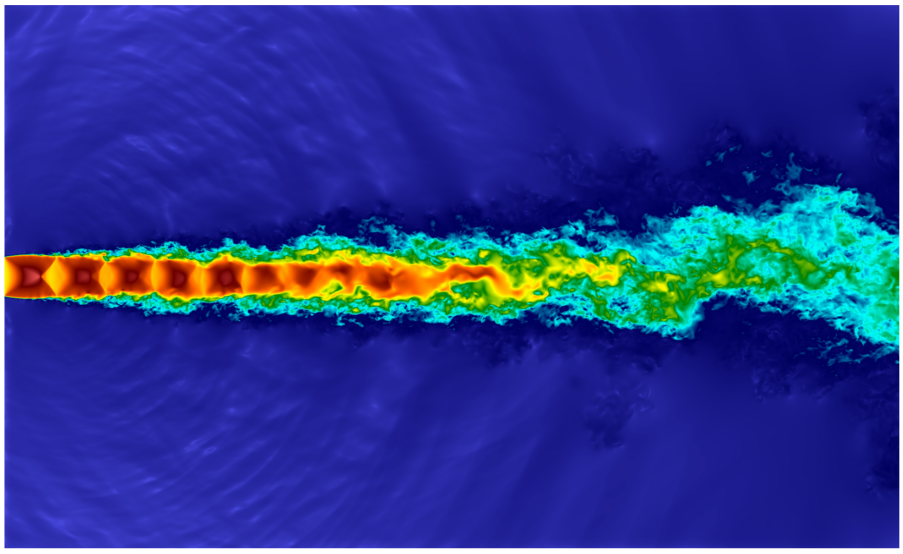


# 1 Introduction

Computational Fluid Dynamics (CFD) is a powerful tool for analyzing and simulating fluid flow problems across various engineering applications. By solving the governing equations of fluid motion numerically, CFD enables engineers to predict flow behavior, pressure distribution, and velocity fields under different conditions.

The flow around a NACA airfoil exhibits different characteristics depending on the Reynolds number, which is a function of the fluid velocity, incidence angle, and kinematic viscosity. At lower Reynolds numbers, the flow remains mostly laminar and symmetric, while at higher Reynolds numbers, vortex shedding occurs, leading to oscillatory wake patterns known as the Kármán vortex street. Understanding these flow features is essential for designing structures that minimize drag and mitigate vibration-induced damage.

# 2 Objectives

The primary objective of this CFD analysis is to investigate the flow characteristics around a NACA airfoil at three different incidence angles: 0°, 4°, 7° and 10°. Specifically, the analysis aims to:

- Compute and visualize the pressure distribution around the profile.
- Analyze and plot the velocity field in the vicinity of the profile.
- Compare how different incidence angles influence the wake region and pressure drop.
- Plot the drag and lift coefficients as a function of the number of iterations and compare the results across different speeds.
- Determine the stall angle

Through this study, we will gain a deeper understanding of fundamental fluid mechanics principles such as boundary layer behavior, pressure drag and vortex shedding. Additionally, the laboratory provides an opportunity to develop proficiency in using CFD software, mesh generation techniques and post-processing tools to interpret simulation results effectively.

# 3 Geometry

For this study, the computational domain consists of a 3D NACA airfoil as shown in Figure 1. The profile is placed in a rectangular flow domain that extends sufficiently upstream and downstream to minimize the influence of boundary effects on the flow solution.

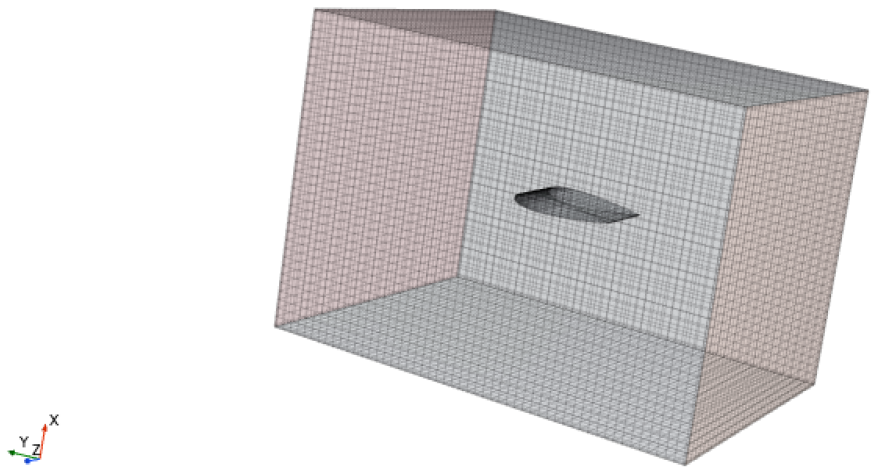



Figure 1: Mesh

The chosen mesh and geometry settings ensure that numerical errors are minimized while keeping computational costs reasonable. The simulations will be conducted using appropriate turbulence models, ensuring accurate flow predictions.

# 4 Results

The results of this CFD analysis provide insight into the flow characteristics around the NACA airfoil. A detailed examination of the pressure and velocity contours will highlight the impact of changing incidence angle (with constant flow's speed equal to $10\,m/s$) on flow separation, wake formation and vortex shedding.

## 4.1 Incidence Angle 0°

The velocity field and contours provide valuable insight into the flow behavior around the NACA airfoil. For technical reasons we decided to create a vertical domain and set an incidence angle equal to 90°, this works in the same way as a 0° incidence angle and an horizontal domain around it. The only thing that change is the reference frame, and so, at the end, the pressure coefficient plot whose depends on the chosen reference frame.

Below are some key observations based on the provided figures:

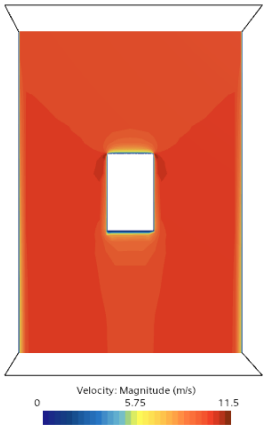



Figure 2: Velocity Contour on a Vertical Plane

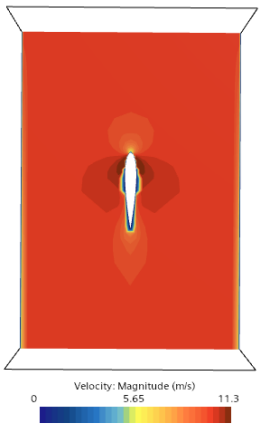



Figure 3: Velocity Contour on a Horizontal Plane

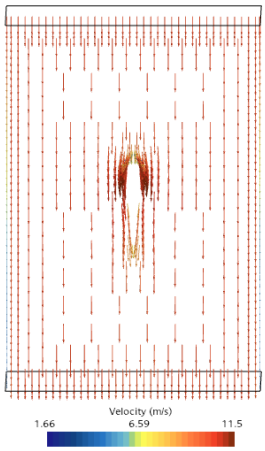



Figure 4: Velocity Field on a Horizontal Plane

Fig.2 shows the contour slice representing the velocity magnitude distribution along a vertical plane passing through the NACA airfoil. The high-velocity region (orange-red) is seen on the sides, where the flow accelerates around the NACA airfoil due to the obstruction of the streamlines. A low-velocity wake region (blue) forms downstream of the NACA airfoil, indicating flow separation. The symmetry in the velocity field suggests a stable flow pattern at this simulation stage.

Fig.3 shows a horizontal slice representing the velocity field at mid-height of the NACA airfoil. The wake behind the profile is evident, characterized by a significant drop in velocity (dark blue region). Flow acceleration around the sides is prominent, confirming the presence of a pressure difference that contributes to drag forces.

Fig.4 effectively illustrates the flow direction and recirculation zones behind the NACA airfoil. Flow stagnation occurs at the front of the NACA airfoil, while acceleration is observed around its sides. The color gradient in the vector field confirms a smooth transition from high velocity upstream to the wake region downstream.

A more detailed analysis of pressure contours would help understand the pressure drag contribution.

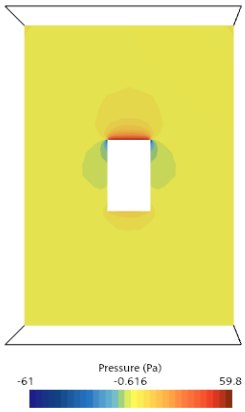



Figure 5: Pressure Contour on a Vertical Plane

Figure 6: Pressure Contour on a Horizontal Plane

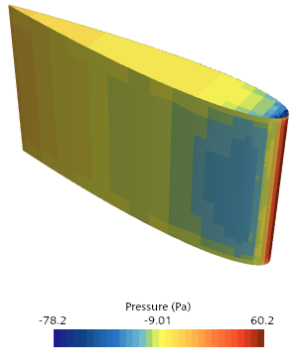



Figure 7: Pressure Contour on the NACA airfoil wall

Fig.5 shows the pressure distribution in a vertical cross-section around the NACA airfoil. The high-pressure region is concentrated near the leading edge, especially at the stagnation point on the front of the airfoil. The pressure then quickly drops on both the upper and lower surfaces due to the acceleration of the airflow. The symmetric distribution suggests the angle of attack is zero, leading to minimal lift generation.

The horizontal cross-sectional view (Figure 6) displays a more pronounced pressure gradient. The highest pressure occurs directly in front of the leading edge, decreasing sharply over the upper surface. This pressure drop is associated with increased velocity (per Bernoulli's principle), especially above the airfoil. The wake behind the airfoil shows a low-pressure region extending downstream, indicating vortex shedding and flow detachment. The pressure field is consistent with subsonic flow behavior over an airfoil at modest angle of attack.

In Fig.7 we can see that the pressure peak at the leading edge transitions to lower values moving toward the trailing edge, reflecting the typical aerodynamic behavior of an airfoil. The top surface (suction side) shows a more significant pressure drop compared to the lower side (pressure side), suggesting that the airfoil is generating lift. The spanwise variation of pressure is minimal, indicating good spanwise flow uniformity and confirming 2D aerodynamic assumptions hold well in this case.

These results are further analyzed in combination with drag/lift and pressure coefficient plots (Fig.8 Fig.9) to assess flow stability and vortex shedding effects.

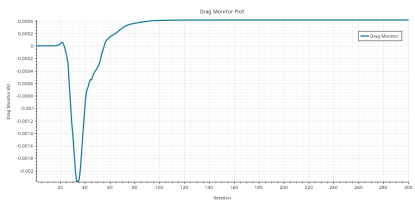



Figure 8: Drag Monitor Plot

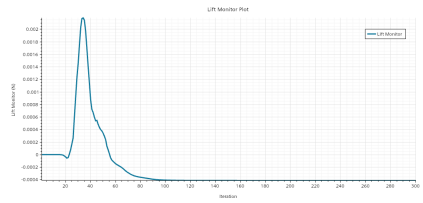



Figure 9: Lift Monitor Plot

The drag of Fig.8 stabilizes after an initial transient phase, indicating that the flow around the NACA airfoil has reached a steady state. The drag coefficient value at steady state can be used to understand the resistance the NACA airfoil experiences due to the fluid flow which in this case is proportional to 0.04% the dynamic pressure.

Similar to the drag coefficient, the lift coefficient shown in Fig.9 stabilizes after an initial transient phase. The steady-state lift coefficient indicates the vertical force acting on the NACA airfoil due to the fluid flow. For a symmetric NACA airfoil, the lift should ideally be zero, but any deviation might be due to the random generation of the mesh elements or other numerical approximations.

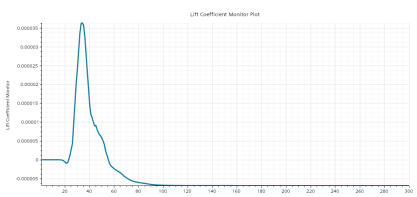



Figure 10: Lift Coefficient Monitor Plot

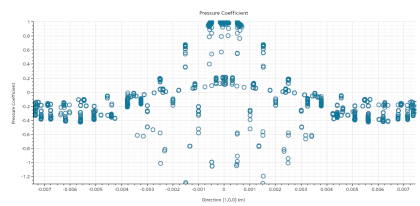



Figure 11: Pressure Coefficient Monitor Plot

The converged value of the lift coefficient in this case is $-6.87976 \cdot 10^{-6}$. Fig. 11 represents the distribution of the pressure coefficient $C_p$ over the airfoil surface along a given direction (X axis in this case!!) The plot shows a typical $C_p$ distribution pattern over an airfoil. The region near $x = 0$ (leading edge) has high values of $C_p$, close to 1, indicating a stagnation point. A sharp drop in $C_p$ right after the leading edge shows high suction on the upper surface due to accelerated flow — this is a characteristic sign of lift generation. Negative $C_p$ values on the upper surface reflect low pressure and high velocity zones (Bernoulli principle), whereas the lower surface has higher pressure values, again confirming lift generation.

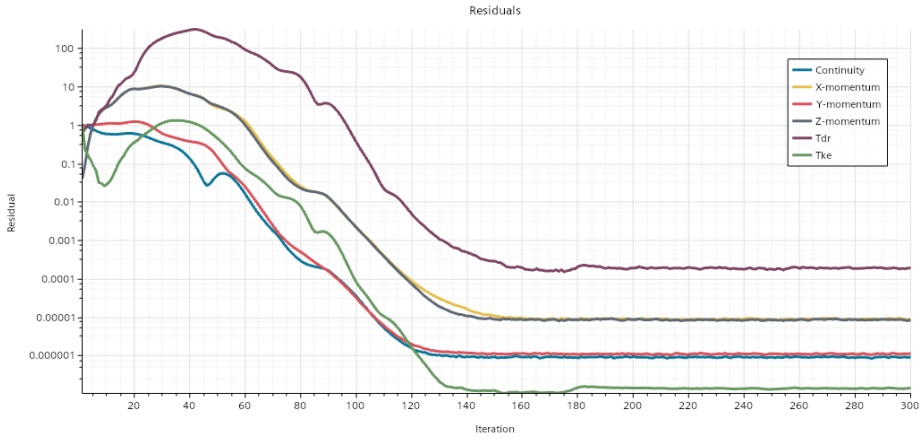



Figure 12: Residuals Plot

Fig. 12 shows the evolution of the residuals for different governing equations (continuity, momentum in x, y, z directions, turbulent dissipation rate (Tdr), and turbulent kinetic energy (Tke)) over the course of 300 solver iterations. All residuals initially increase and then decrease steadily, which is typical behavior as the solver approaches a solution from the initial guess. Around iteration 100 to 150, most of the residuals drop below the typical convergence threshold (usually around $10^{-3}$ to $10^{-5}$), particularly the momentum and turbulence variables. However, the continuity residual and Z-momentum stabilize but don't reach as low values as the others. This may indicate minor numerical oscillations or some lingering instability in that component of the solution (often due to mesh quality, boundary conditions, or flow complexity). Overall, the solution shows good convergence behavior, and the flat residual lines after 200 iterations confirm that the solution is stable and has reached a steady state.

## 4.2 Incidence Angle 4°

Changing the incidence angle from 0° to 4° in the CFD simulation significantly impact the pressure distribution, velocity field, and aerodynamic coefficients.

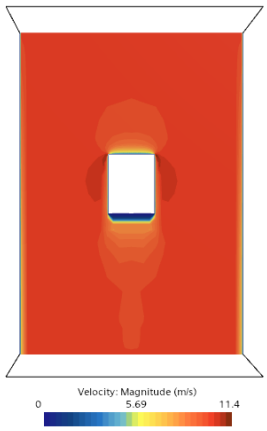



Figure 13: Velocity Contour on a Vertical Plane

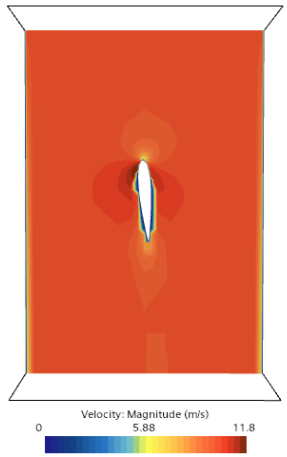



Figure 14: Velocity Contour on a Horizontal Plane

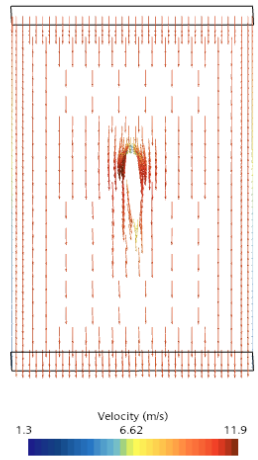



Figure 15: Velocity Field on a Horizontal Plane

This vertical cross-section shows an evident asymmetry compared to the original 0° case. The deflection in the velocity contours, especially near the trailing edge of the profile, indicates the generation of lift due to the 4° angle of attack. The region of low velocity beneath the profile suggests increased pressure on the lower surface, which is consistent with the pressure-lift relationship in airfoils. Wake elongation is also visible, reflecting altered flow detachment behavior.

The horizontal plane further confirms the shift in flow structure. The symmetry from the 0° case is now broken, and we can observe a stronger velocity gradient around the profile, especially near the leading edge. Flow acceleration over the upper surface becomes more pronounced due to the angle change, contributing to the expected increase in lift and possibly a slight increase in drag due to induced flow curvature.

The velocity vector field offers a clear visualization of flow direction deviation. The vectors downstream of the airfoil now show vertical components, confirming the generation of lift (via downward momentum in the wake, in accordance with Newton's third law). There is also a visible recirculation zone behind the airfoil, suggesting that separation may have slightly intensified, especially near the trailing edge or at the suction side.

Figure 16: Pressure Contour on a Vertical Plane

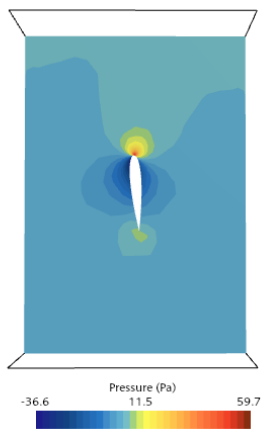



Figure 17: Pressure Contour on a Horizontal Plane

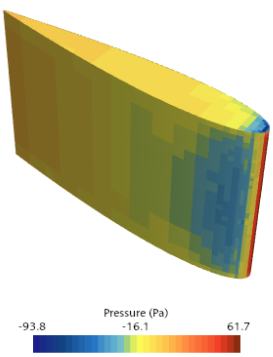



Figure 18: Pressure Contour on the NACA airfoil wall

Compared to the symmetric pressure distribution at 0° angle of attack, this vertical slice now shows a more pronounced asymmetry. There's a clear region of higher pressure beneath the profile and a lower pressure region above it. This pressure differential is a direct indicator of positive lift generation. Additionally, the contour distortion downstream suggests a modified wake region, likely thicker and slightly more turbulent, corresponding to the flow deflection and lift.

The horizontal plane also demonstrates a shift in pressure distribution. We observe higher pressure gradients on the lower side of the airfoil and more intense low-pressure zones on the upper surface. This pattern confirms the increased aerodynamic loading due to the new angle of incidence. Also, the shape of the pressure field behind the profile is stretched further, indicating more intense wake effects and possibly early signs of separation depending on the flow regime and Reynolds number.

On the airfoil surface itself, the pressure field reveals a stronger suction peak near the leading edge on the upper surface, which is typical at positive angles of attack. The lower surface displays increased pressure as expected, enhancing the lift force. The pressure distribution is now clearly skewed, showing the aerodynamic asymmetry introduced by the 4° incidence. These features are consistent with typical NACA profile behavior in subsonic flow at low angles of attack.

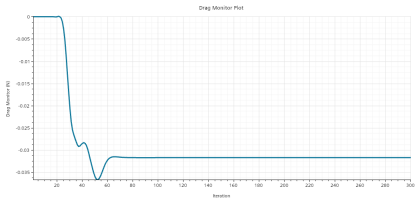



Figure 19: Drag Monitor Plot

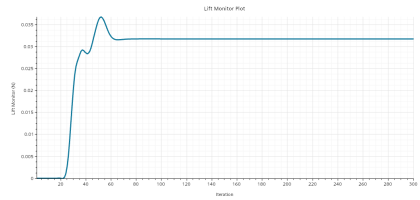



Figure 20: Lift Monitor Plot

At 0°, lift was nearly negligible due to symmetric flow. With 4°, the airfoil generates a stronger pressure difference between its upper and lower surfaces, evident in the pressure contours. This leads to positive lift, in fact now lift is significantly greater. An increase in drag force is also visible. The profile now experiences both form drag (due to asymmetrical shape and pressure distribution) and induced drag (resulting from lift generation).

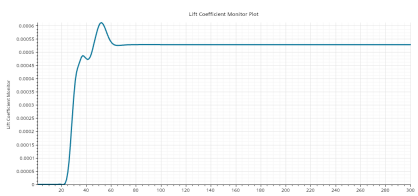



Figure 21: Lift Coefficient Monitor Plot

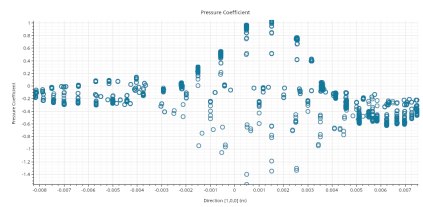



Figure 22: Pressure Coefficient Monitor Plot

The converged value of the lift coefficient in this case is $5.2846 \cdot 10^{-4}$. The $C_p$ distribution is no longer symmetric as it was at 0°. The upper surface shows stronger negative $C_p$ values (indicating suction), especially near the leading edge — a classic sign of lift. The lower surface shows more positive $C_p$ values, indicating increased pressure. This asymmetry confirms lift generation and the onset of pressure gradients caused by the incidence. Some scattered data may suggest local flow instabilities or mesh/coarse sampling, but overall the trend is clear and consistent with classical thin airfoil theory.

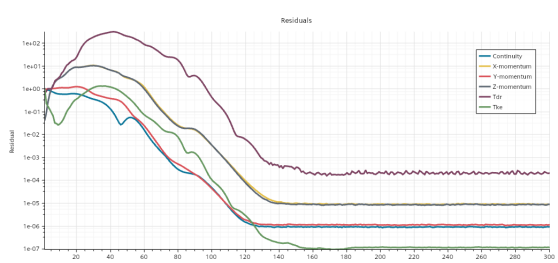



Figure 23: Residuals Plot

## 4.3 Incidence Angle 7° and 10°

Since the variations from the previous case only increase the magnitude forces without any big difference in the behavior, it's become trivial to show again all plots. Let's do indeed a comparison among lift coefficients in order to determine the stall angle which is the angle at which the lift coefficient peaks.

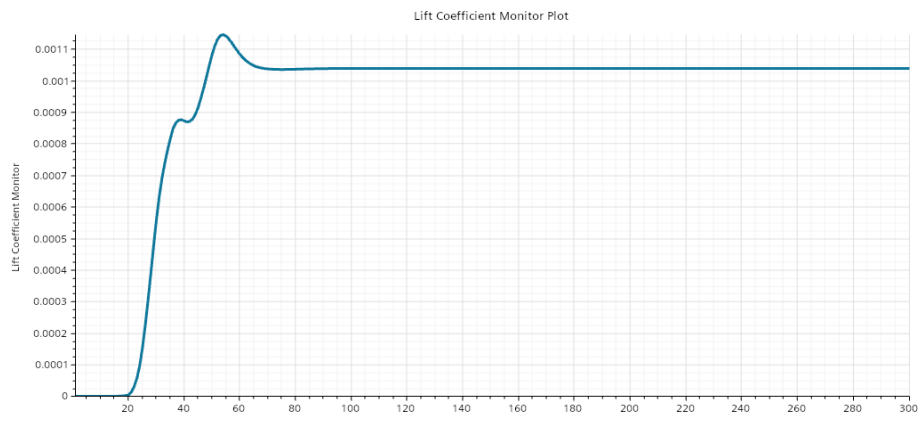



Figure 24: Lift Coefficient Monitor Plot with 7° angle of attack

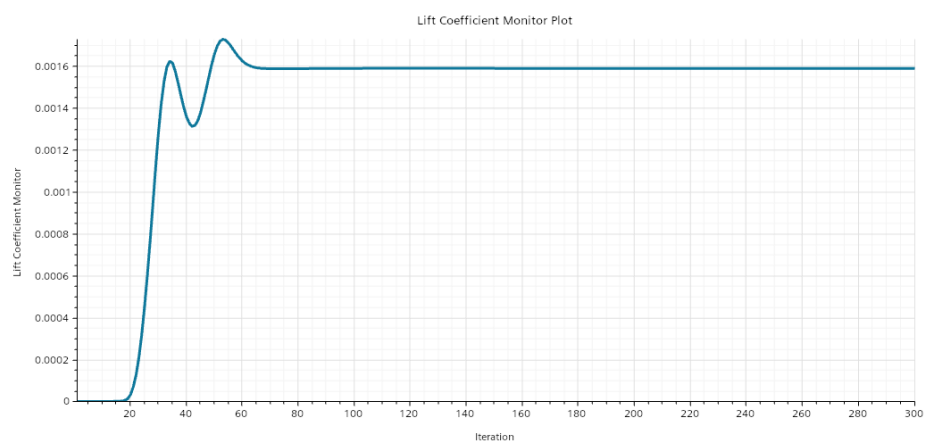
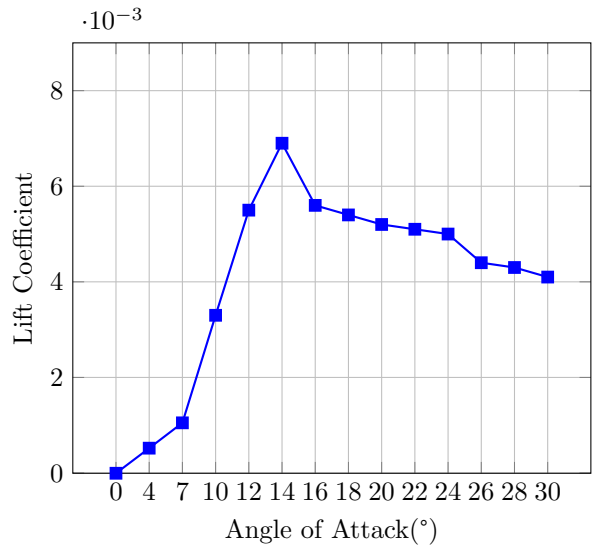



Figure 25: Lift Coefficient Monitor Plot with 10° angle of attack

We repeated the same procedure with the following angles: 12°, 14°, 16°, 18°, 20°, 22°, 24°, 26°, 28° 30°.




# 5 Conclusions

In this study, a comprehensive aerodynamic analysis of a NACA airfoil profile was performed under varying angles of attack at constant flow speed. At 0° incidence, the profile demonstrated low lift and symmetric flow behavior, with minimal drag and stable pressure and velocity contours. Increasing the angle to 4° and 7° led to a progressive rise in lift, while the flow remained attached and contours began to show greater asymmetry, consistent with typical pre-stall aerodynamic behavior. Further analysis revealed that the lift coefficient continued increasing with angle until reaching a stall point at approximately 14°, beyond which the aerodynamic performance degraded. This stall angle was confirmed by a sudden change in the lift trend and supporting flow visualizations that showed flow separation. Overall, the results validated the expected aerodynamic performance of the airfoil and highlighted 14° as the critical stall angle for this configuration.